

TCS3351 CRYPTOGRAPHY AND DATA SECURITY
TUTORIAL 2
MATHEMATICAL BACKGROUND – I

1. a) Which of the following relations are true and which are false?

(i) $5 \mid 26$

(ii) $3 \mid 123$

(iii) $27 \nmid 127$

(iv) $15 \nmid 21$

Can 5 divide 26? $\Rightarrow$ No. $\Rightarrow$ false

$3 \mid 123 = 41$ true

$27 \nmid 127$ true

$15 \nmid 21$ true

- b) Find the results of the following operations.

(i) $-78 \bmod 13$

(ii) $0 \bmod 13$

(iii) $150 \bmod 15$

(iv) $-66 \bmod 33$

All zero

Step 1: Divide the number by the mod value. Remove the digit, leave the fraction. Multiple the fraction with the mod value.

$-78/13 = -6.0$,

$-78 \bmod 13 = [(-13) \times 6] \bmod 13$
 $= 0.$

$0 \bmod 13 = 0$

$150 \bmod 15 = (10 \times 15) \bmod 15 = 10 \times (15 \bmod 15) = 0$

$-66 \bmod 33 = -2 \times 33 \bmod 33 = -2 \times 0 = 0$

Let do

$97 \bmod 26 =$

$-97 \bmod 26 =$

Divide the number by the mod value. Remove the digit, leave the fraction. Multiple the fraction with the mod value.

$-97/26 = -3.7308$

$-3.7308 + 3 = -0.7308$

$-0.7308 \times 26 = -19$

$-97 \bmod 26 = -19 \bmod 26 = (-19 + 26) \bmod 26 = 7$

Step 2: If the value is –ve, then add the modulus value into it.

2. a) Using the Euclidean algorithm, find the greatest common divisor of the pairs on integers, 88 and 220.

- b) Using the extended Euclidean algorithm, find the greatest common divisor of the pairs (4,7) and the value of s and t.

$\text{Gcd}(88, 220) =$

$220 = 88(2) + 44$

$88 = 44(2) + 0$; remainder = 0, so GCD = 44

$\text{Gcd}(4, 7) =$

$$\begin{aligned}
7 &= 4(1) + 3 \\
4 &= 3(1) + 1 \\
3 &= 1(3) + 0 ; \text{ remainder} = 0, \text{ GCD} = 1
\end{aligned}$$

$$\begin{aligned}
&\text{Find gcd}(36, 1180) = 4 \\
1180 &= 36(32) + 28 \\
36 &= 28(1) + 8 \\
28 &= 8(3) + 4 \\
8 &= 4(2) + 0; \text{ remainder} = 0, \text{ GCD} = 4.
\end{aligned}$$

$$\begin{aligned}
&\text{Find gcd}(36, 1182) \\
1182 &= 36(32) + 30 \\
36 &= 30(1) + 6 \\
30 &= 6(5) + 0. \text{ Remainder} = 0, \text{ GCD} = 6.
\end{aligned}$$

3. Perform the following operations using reduction first.

- (i) $(4223 + 17323) \bmod 10$
- (ii) $(2468 - 43) \bmod 15$
- (iii) $(144 \times 34) \bmod 12$
- (iv) $(221 \times 23) \bmod 22$

$$\begin{aligned}
&(4223 + 17323) \bmod 10 \\
&= 422\textbf{3} \bmod 10 + 1732\textbf{3} \bmod 10 \\
&= 3 \bmod 10 + 3 \bmod 10 \\
&= 6
\end{aligned}$$

$$\begin{aligned}
&(2468 - 43) \bmod 15 \\
&= 2468 \bmod 15 - 43 \bmod 15 \\
&= (8 - 13) \bmod 15 \\
&= -5 \bmod 15 \\
&= -5 \bmod 15 = (-5 + 15) \bmod 15 = 10 \bmod 15 = 10
\end{aligned}$$

$$\begin{aligned}
&(144 \times 34) \bmod 12 \\
&= 144 \bmod 12 \times 34 \bmod 12 \\
&= 0 \times 34 \bmod 12 \\
&= 0
\end{aligned}$$

$$\begin{aligned}
&(221 \times 23) \bmod 22 \\
&= 221 \bmod 22 \times 23 \bmod 22 \\
&= 1 \times 1 \\
&= 1
\end{aligned}$$

4. Check whether the multiplicative inverse exists for each of the following integers and if exists, find it using the extended Euclidean algorithm.

- (i) 61 in $\mathbf{Z}_{90}$
- (ii) 132 in $\mathbf{Z}_{180}$

multiplicative inverse exists when $\text{gcd}(x, y) = 1$; $\text{gcd}(61, 90) = 1$ (true/false)

$a = qn + r$; formula

$$\mathbf{ax + by = 1;}$$

a = encryption key.

$x = a^{-1}$. decryption key. y the modulus value.

Where a. $a^{-1} == 1$

$Z_{90} = \{0, 1, 2, 3, \dots, 89\}$

Gcd(61, 90) = 1 Ext Euclidean algo $90 = 61 \cdot 1 + 29 \Rightarrow 29 = 90 - 61 \dots \text{eq 1}$ $61 = 29 \cdot 2 + 3 \Rightarrow 3 = 61 - 29 \cdot 2 \dots \text{eq 2}$ $29 = 3 \cdot 9 + 2 \Rightarrow 2 = 29 - 3 \cdot 9 \dots \text{eq 3}$ $3 = 2 \cdot 1 + 1 \Rightarrow 1 = 3 - 2 \cdot 1 \dots \text{eq 4}$ $2 = 1 \cdot 2 + 0$ Gcd(61,90) = 1 , thus inverse multiplication exist	Gcd(132,180) $180 = 132(1) + 48$ $132 = 48(2) + 36$ $48 = 36(1) + 12$ $36 = 12(3) + 0$; gcd(132,180) = 12 $\neq 1$ Inverse multiplication does not exist!
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From eq 4 to eq 1 in reverse order

From eq4,

$$1 = 3 - 2 \cdot 1$$

Substitute 2 in eq3,

$$1 = 3 - (29 - 3 \cdot 9) = 10(3) - 29$$

Substitute 3 in eq2,

$$1 = 10(61 - 29 \cdot 2) - 29 = 10(61) - 20(29) - 29$$

$$1 = 10(61) - 21(29)$$

Substitute 29 in eq1

$$1 = 10(61) - 21(90 - 61)$$

$$= 10(61) + 21(61) - 21(90)$$

$$1 = \mathbf{31}_{(61)} - 21(90)$$

Thus, the inverse of 61 is 31.

Find the inverse of 56 in Z_{97}

$$1 = ax + by, a = 61 \text{ (key)}, b = 90, y = -21; I$$

$$61 \cdot (61^{-1}) \bmod 90 = 1$$

$$\text{Proof, } 61 \cdot 31 = 1891.$$

$$1891 \bmod 90 = (1890 + 1) \bmod 90$$

5. Let us assign numeric values to the uppercase alphabet (A=0, B=1, ..., Z=25). Answer the following using modulo 26.

(i) $(A+N) \bmod 26$

(ii) $(A+6) \bmod 26$

(ii) $(Y-5) \bmod 26$

(iv) $(C-10) \bmod 26$

a	B	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

$(A+N) \bmod 26$ $= (0 + N) \bmod 26$ $= N \bmod 26$ $= N (13)$	$(A+6) \bmod 26 = 6 (G)$	$(Y-5) \bmod 26$ $= 24 - 5 \bmod 26 = 19 (t)$
$(C-10) \bmod 26$ $= (2-10 + 26) \bmod 26$		

=18 mod 26 =18 (S)		
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References:

- 1) Behrouz A. Forouzan, Debdeep Mukhopadhyay, "Cryptography and Network Security", Third Edition, McGraw-Hill, 2015.
- 2) William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2017.

Extra Exercises:

1.9. Use the Euclidean algorithm to compute the following greatest common divisors.

- (a) $\text{gcd}(291, 252)$.
- (b) $\text{gcd}(16261, 85652)$.
- (c) $\text{gcd}(139024789, 93278890)$.
- (d) $\text{gcd}(16534528044, 8332745927)$.

1.17. Do the following modular computations. In each case, fill in the box with an integer between 0 and $m - 1$, where m is the modulus.

(a) $347 + 513 \equiv \boxed{} \pmod{763}$.

(b) $3274 + 1238 + 7231 + 6437 \equiv \boxed{} \pmod{9254}$.

(c) $153 \cdot 287 \equiv \boxed{} \pmod{353}$.

(d) $357 \cdot 862 \cdot 193 \equiv \boxed{} \pmod{943}$.

(e) $5327 \cdot 6135 \cdot 7139 \cdot 2187 \cdot 5219 \cdot 1873 \equiv \boxed{} \pmod{8157}$.

(*Hint.* After each multiplication, reduce modulo 8157 before doing the next multiplication.)

(f) $137^2 \equiv \boxed{} \pmod{327}$.

(g) $373^6 \equiv \boxed{} \pmod{581}$.

(h) $23^3 \cdot 19^5 \cdot 11^4 \equiv \boxed{} \pmod{97}$.

1.18. Find all values of x between 0 and $m - 1$ that are solutions of the following congruences. (*Hint.* If you can't figure out a clever way to find the solution(s), you can just substitute each value $x = 1, x = 2, \dots, x = m - 1$ and see which ones work.)

(a) $x + 17 \equiv 23 \pmod{37}$.

(b) $x + 42 \equiv 19 \pmod{51}$.

1.32. For each of the following primes p and numbers a , compute $a^{-1} \pmod{p}$

(a) $p = 47$ and $a = 11$.

(b) $p = 587$ and $a = 345$.

(c) $p = 104801$ and $a = 78467$.

References:

- 1) Jeffrey Hoffstein, Jill Pipher and Joseph H. Silverman, "An Introduction to Mathematica; Cryptography", Second Edition, Springer, 2014.